

Q.1

Which call will print exactly: 1 2 3 4 5

```
def count_up(start, stop, step=1):  
    i = start  
    while i <= stop:  
        print(i, end=' ')  
        i += step
```

`count_up(1, 5)`

`count_up(0, 5)`

`count_up(1, 4)`

`count_up(1, 5, 2)`

Q.2

What is the output of the following code?

```
def mystery(n):  
    total = 0  
    for i in range(1, n):  
        if i % 2 == 0:  
            total += i  
    return total  
  
print(mystery(8))
```

10

12

16

20

Q.3

What does this function return when called as shown?

```
def flatten(lst, result=None):
    if result is None:
        result = []
    for item in lst:
        if isinstance(item, list):
            flatten(item, result)
        else:
            result.append(item)
    return result

print(flatten([1, [2, [3, 4]], 5]))
```

[1, 2, 3, 4, 5]

[1, [2, 3, 4], 5]

[1, 2, [3, 4], 5]

Error

Q.4

What is printed?

```
funcs = []
for x in range(3):
    funcs.append(lambda: x)

print([f() for f in funcs])
```

[0, 1, 2]

[0, 0, 0]

[2, 2, 2]

Error

Q.5

What is the exact output of this code?

```
def f(x, y=2):  
    return x ** y  
  
result = [f(i) for i in range(1, 5)]  
print(result)
```

[1, 4, 9, 16]

[1, 2, 3, 4]

[2, 4, 6, 8]

[1, 4, 8, 16]

Q.6

What is printed?

```
def gen(n):  
    i = 0  
    while i < n:  
        val = yield i  
        i += (val or 1)  
  
g = gen(10)  
print(next(g))  
print(g.send(3))  
print(next(g))
```

0 3 4

0 1 4

0 3 6

0 3 7

Q.7

What is the exact output?

```
def counter(start=0):  
    count = [start]  
    def increment(n=1):  
        count[0] += n  
        return count[0]  
    return increment  
  
c = counter(10)  
print(c())  
print(c(5))  
print(c())
```

11 16 17

1 6 7

10 15 16

11 15 16

Q.8

This code should print the **running total** of CV scores after each fold. Find the bug and fix it.

```
python  
  
cv_scores = [0.82, 0.79, 0.84, 0.81, 0.83]  
total = 0  
  
for i, score in enumerate(cv_scores):  
    total = score  
    print(f"After Fold {i+1}: running total = {total:.4f}")
```

Q.9

This code should count how many values in the dataset are `None` (missing). Find the bug and fix it.

```
python

data = [25, None, 0.8234, None, 'Age', True, None, 891]
missing = 0

for value in data:
    if value == None:
        missing += 1

print(f"Missing values: {missing}")
```

Q.10

Q10: Find the bug

```
python

results = {'train': 0.9123, 'test': 0.8234, 'gap': 0.0889}

print(results['Train'])
print(results.get('gap', 0))
```